

HemoFlow AI — Features & Functionalities

1. Hospital Request Management

Hospitals can raise blood requests by selecting:

- Blood group
- Blood component
- Required units
- Priority level
- Request reason

The system centralizes emergency blood coordination across multiple hospitals.

2. Intelligent Approval Center

The Approval Center allows hospitals to:

- Approve requests
- Reject requests
- Escalate requests

Approval Logic

- If approved → transfer process starts
- If rejected → request disappears only for that hospital
- Other hospitals can still approve
- If escalated → request becomes visible to all hospitals

This creates decentralized and intelligent hospital coordination.

3. Human-in-the-Loop (HITL) Workflow

Critical decisions are not fully automated.

Hospital staff remain in control while AI provides recommendations.

This ensures:

- safety
- trust
- operational accountability

4. AI Insights Panel

AI Agents analyze operational hospital data and provide:

- blood shortage alerts
- urgent request prioritization
- transfer recommendations
- inventory risk analysis
- operational bottleneck detection

The AI acts as an operational intelligence assistant for hospitals.

5. Transfer Tracking System

Once a request is approved:

- Unique Transfer ID is generated
- Transfer lifecycle begins

Hospitals can track:

- donor hospital
- receiving hospital
- transfer status
- transfer timeline

Transfer Status Timeline

- Request Raised
 - Approved
 - Blood Prepared
 - In Transit
 - Delivered
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6. Multi-Hospital Coordination

HemoFlow AI connects hospitals into a collaborative ecosystem where hospitals can support each other during emergencies.

This improves:

- resource sharing
- emergency response speed

- blood availability visibility
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7. Inventory Management

Hospitals can monitor:

- available blood units
- blood component availability
- inventory shortages
- transfer history

This helps prevent critical shortages.

8. Agentic AI Operational Analysis

AI Agents dynamically:

- fetch operational records
- analyze transfer and inventory data
- generate concise operational recommendations

The AI system supports faster and smarter decision-making.

9. Workflow Orchestration

The system uses UiPath workflows to:

- manage approvals
- trigger AI analysis
- coordinate transfers
- update operational records

This reduces manual operational workload.

10. Scalable Healthcare Automation Architecture

HemoFlow AI is designed as a scalable healthcare coordination platform using:

- UiPath Apps
- AI Agents
- Data Service
- Human-in-the-Loop automation

The architecture supports future expansion into:

- predictive analytics
- donor systems
- real-time logistics
- smart emergency healthcare networks